Auction with Secret Reserve Price

Contents

1 Model Outline

- One seller, N bidders.
- Seller's cost is either c_L or c_H with $\Pr(c_L) = \psi$.
- Buyer's value is i.i.d. distributed according to some F supported on $[c_L, 1]$, with density f.
- **First stage:** an English auction with initially hidden reserve price, which is revealed conditional on being met.
- **Second stage:** if the English auction fails, one bidder is randomly drawn to make a TIOLI (take-it-or-leave-it) offer to the seller. An individual bidder is drawn with probability $\delta \le 1/N$.

Notation. $\underline{r}$ and $\bar{r}$: endogenous lower and upper support of the low-cost seller's mixed strategy (chosen in equilibrium, not drawn); r_H: high-cost seller's reserve; $\rho \in \Delta[\underline{r}, \bar{r}]$: cdf of the reserve r actually played on the path; β: symmetric stage-1 bidding strategy; $\tilde{\psi}$: belief that the seller is low-cost.

2 Preliminary Analysis

2.1 Second-stage belief and offer

If the chosen buyer has a belief $\tilde{\psi}$, he offers price c_H if and only if

$$v - c_H \ge \tilde{\psi}(v - c_L) \quad \Longleftrightarrow \quad v \ge \frac{c_H - \tilde{\psi} c_L}{1 - \tilde{\psi}}. \tag{1}$$

2.2 Equilibrium structure

Consider an equilibrium where the high-cost seller plays a pure strategy r_H and the low-cost seller plays a mixed strategy denoted by $\rho \in \Delta[\underline{r}, \bar{r}]$.

Claim 1. $r_H > \bar{r}$.

Proof in Section 6. For any increasing bidding function, the optimal reserve price for a high-cost seller is strictly above that of a low-cost seller.

2.3 High-cost seller

When the bid exceeds $\bar{r}$ but the reserve is not met, bidders believe the seller is high-cost and bid according to $\beta(v) = v - \delta(v - c_H)$. Once a bidder has $\beta(v) \ge r_H$, the reserve is announced as met and people bid their own values. The high-cost seller's profit $\Pi_H(r_H)$ is

$$\Pi_H(r_H) = \int_{\beta^{-1}(r_H)}^{1} \int_{c_L}^{v} \big(\max\{r_H, s\} - c_H\big)\, d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N. \tag{2}$$

Simplifying,

$$
\begin{aligned}
\Pi_H(r_H) &= N\int_{\beta^{-1}(r_H)}^{1}\Big[(r_H-c_H)F(r_H)^{N-1}+\int_{r_H}^{v}(s-c_H)\,dF(s)^{N-1}\Big]\,dF(v)\\
&= N\Bigg[(r_H-c_H)F(r_H)^{N-1}\big(1-F(\beta^{-1}(r_H))\big)+\int_{r_H}^{1}(s-c_H)\,dF(s)^{N-1}\\
&\qquad -F(\beta^{-1}(r_H))\int_{r_H}^{\beta^{-1}(r_H)}(s-c_H)\,dF(s)^{N-1}-\int_{\beta^{-1}(r_H)}^{1}F(v)(v-c_H)\,dF(v)^{N-1}\Bigg].
\end{aligned}
\tag{3}
$$

The first-order condition is

$$
r_H-c_H=(1-\delta)\frac{1-F\left(\dfrac{r_H-\delta c_H}{1-\delta}\right)}{f\left(\dfrac{r_H-\delta c_H}{1-\delta}\right)}-\frac{1}{F(r_H)^{N-1}}\int_{r_H}^{(r_H-\delta c_H)/(1-\delta)}(v-c_H)\,dF(v)^{N-1}. \tag{4}
$$

2.4 Belief after a failed English auction

If the English auction fails with final bid b, the buyer's belief is

$$
\tilde{\psi}=\frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}. \tag{5}
$$

3 Bidding in the First-Stage English Auction

Suppose a bidder with value v decides his bid by comparing $v-b$ against

$$
\delta\max\left\{v-c_H,\ \frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}(v-c_L),\ 0\right\}. \tag{6}
$$

His bid thus satisfies

$$
b=v-\delta\max\left\{v-c_H,\ \frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}(v-c_L),\ 0\right\}. \tag{7}
$$

The right-hand side is weakly decreasing in b and bounded between $[v-\delta\max\{v-c_H,\psi(v-c_L)\},\ v-\delta\max\{v-c_H,0\}]$. The bidding strategy β given ρ is thus

$$
\beta(v)=\begin{cases}
v-\delta\psi(v-c_L) & \text{if } v\le \dfrac{\underline{r}-\delta\psi c_L}{1-\delta\psi},\\
h^{-1}(v) & \text{if } v\in\left(\dfrac{\underline{r}-\delta\psi c_L}{1-\delta\psi},\ v^*\right),\\
v-\delta(v-c_H) & \text{if } v\ge v^*,
\end{cases} \tag{8}
$$

where

$$
h(b)\equiv\frac{(1-\psi\rho(b))b-\delta\psi(1-\rho(b))c_L}{1-\delta\psi-(1-\delta)\psi\rho(b)}, \tag{9}
$$

i.e. $h(b)$ solves $v - b = \delta\tilde{\psi}(b)(v - c_L)$ with $\tilde{\psi}(b) = \psi(1 - \rho(b))/(1 - \psi\rho(b))$ from (5), and $v^* \in \big[(\underline{r} - \delta\psi c_L)/(1 - \delta\psi),\ (c_H - \psi c_L)/(1 - \psi)\big]$ is the solution to

$$v - c_H = \frac{\psi\big(1 - \rho((1-\delta)v + \delta c_H)\big)}{1 - \psi\rho((1-\delta)v + \delta c_H)}(v - c_L) \quad\iff\quad \rho\big((1-\delta)v + \delta c_H\big) = \frac{\psi(v - c_L) - (v - c_H)}{\psi(c_H - c_L)}. \tag{10}$$

4 Low-Cost Seller Profit

It suffices to focus on the low-type seller's profit. Suppose it sets reserve price $\underline{r}$.

4.1 When the auction fails

The English auction *fails* when every bidder stops before the reserve $\underline{r}$ is met, so the sale moves to stage 2. With symmetric bidding β, that requires all N values to satisfy $\beta(v) \le \underline{r}$, which occurs with probability

$$\Pr(\text{fail}) = F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^N. \tag{11}$$

The seller may still earn revenue in stage 2. One bidder is drawn to make a take-it-or-leave-it offer with probability δ; across N bidders this contributes a factor δN. The selected buyer is unsure of the seller's cost and, by (1), offers c_H only if $v - c_H$ beats the expected gain from dealing with a low-cost seller. In the expression below we use the cutoff implied by prior belief ψ, $v \ge (c_H - \psi c_L)/(1 - \psi)$. If the seller is in fact low-cost, serving the good still costs only c_L, so each accepted offer at c_H yields profit $c_H - c_L$.

Conditional on stage 1 failure, all values are relatively low—the maximum is at most $\beta^{-1}(\underline{r})$. A selected bidder offers c_H only if his own value exceeds the cutoff above. Profit in this event is therefore

$$\Pi_{\text{fail}} = \delta N(c_H - c_L) \times \Pr\left(v_1 \ge \frac{c_H - \psi c_L}{1 - \psi} \;\middle|\; v_{\max} \le \beta^{-1}(\underline{r})\right), \tag{12}$$

Write

$$v_\psi \equiv \frac{c_H - \psi c_L}{1 - \psi}, \qquad z \equiv \beta^{-1}(\underline{r}) = \frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}.$$

By (1), a selected bidder offers c_H only if $v \ge v_\psi$. Conditional on $v_{\max} \le z$, this requires $z \ge v_\psi$, so (12) is positive if and only if

$$z > v_\psi \quad\iff\quad F(z) > F(v_\psi) \quad\iff\quad 1 - \frac{F(v_\psi)}{F(z)} > 0. \tag{13}$$

The reserve threshold that makes $z = v_\psi$ is

$$\underline{r}_0 \equiv \delta\psi c_L + (1 - \delta\psi)\frac{c_H - \psi c_L}{1 - \psi} = \frac{(1 - \delta\psi)c_H - \psi(1 - \delta)c_L}{1 - \psi}, \qquad \text{so} \quad \underline{r} > \underline{r}_0 \quad\iff\quad z > v_\psi. \tag{14}$$

When $\underline{r} > \underline{r}_0$, failure leaves some values in $[v_\psi, z]$, so a selected bidder may have $v \ge v_\psi$ and offer c_H, and stage-2 revenue is possible.[1] Thus $\Pi_{\text{fail}} = 0$ for $\underline{r} \le \underline{r}_0$. For $\underline{r} > \underline{r}_0$, i.i.d. values and (11)

[1] The cutoff $\underline{r}_0$ need not lie below the top of the value support. If buyers place sufficiently high prior probability on the low-cost type, their cutoff for offering c_H is so high that even the largest feasible failed-auction value may not trigger a c_H offer. Indeed, $\underline{r}_0 < 1$ if and only if $\psi < (1 - c_H)/\big(1 - c_L - \delta(c_H - c_L)\big)$; if not, then $\underline{r}_0 \ge 1$ and $\Pi_{\text{fail}} = 0$ for every feasible reserve $\underline{r} \le 1$.

give

$$\Pi_{\text{fail}} = \delta N(c_H - c_L)\ \Pr(\text{fail}) \left(1 - \frac{F(v_\psi)}{F(z)}\right) = \delta N(c_H - c_L)\left(F(z) - F(v_\psi)\right) F(z)^{N-1}. \tag{15}$$

Raising $\underline{r}$ makes stage 1 failure more likely because $F(z)^N$ increases, and for $\underline{r} > \underline{r}_0$ it also raises the probability in (12) that a selected bidder offers c_H.

4.2 When the auction succeeds

The auction *succeeds* when at least one bidder is willing to push the bid above $\underline{r}$, so the reserve is met and the object is sold in stage 1. This requires $v_{\max} > \beta^{-1}(\underline{r})$, which occurs with probability $1 - F\big((\underline{r} - \delta\psi c_L)/(1-\delta\psi)\big)^N$. Once the reserve is met, bidders bid truthfully in the English auction. The bidder with the highest value v wins. The runner-up has the second-highest value s; the winner pays $\max\{\underline{r}, s\}$ (the reserve if $s < \underline{r}$, otherwise s). The seller's margin is therefore $\max\{\underline{r}, s\} - c_L$. On the low bidding branch, $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1-\delta\psi) > \underline{r}$ when $\underline{r} > c_L$, so every winning type satisfies $v > \underline{r}$ and the payment splits into two cases: the winner pays $\underline{r}$ when $s < \underline{r}$, and pays s when $s > \underline{r}$. Averaging over the winner's type v and the second-order statistic s gives

$$\Pi_{\text{succ}} = \int_{\beta^{-1}(\underline{r})}^{1} \int_{c_L}^{v} \big(\max\{\underline{r}, s\} - c_L\big)\, d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N, \tag{16}$$

where $d(F(s)/F(v))^{N-1}$ is the distribution of s conditional on $v_{\max} = v$, and $dF(v)^N = NF(v)^{N-1}f(v)\,dv$ is the density of the maximum. Integrating by parts yields the equivalent form

$$\Pi_{\text{succ}} = N\left\{(\underline{r} - c_L)F(\underline{r})^{N-1}\big(1 - F(\beta^{-1}(\underline{r}))\big) + \int_{\beta^{-1}(\underline{r})}^{1} \int_{\underline{r}}^{v} (s - c_L)\, dF(s)^{N-1}\, dF(v)\right\}. \tag{17}$$

The first term collects states where the runner-up's value is below $\underline{r}$, so the sale price equals the reserve. The double integral collects states where $s > \underline{r}$, so the winner pays the runner-up's value s (the usual second-price payment in an English auction).

4.3 Total profit and $\underline{r}^*$

Taken together, the low-type's profit is, with $[x]_+ \equiv \max\{x, 0\}$,

$$\begin{aligned}\Pi(\underline{r}) \equiv{}& \delta N(c_H - c_L)\left[F(z) - F(v_\psi)\right]_+ F(z)^{N-1} \\ &+ N\left\{(\underline{r} - c_L)F(\underline{r})^{N-1}\big(1 - F(\beta^{-1}(\underline{r}))\big) + \int_{\beta^{-1}(\underline{r})}^{1} \int_{\underline{r}}^{v} (s - c_L)\, dF(s)^{N-1}\, dF(v)\right\}.\end{aligned} \tag{18}$$

The formula $\Pi(r)$ attributes stage-2 payoffs using the prior ψ on every failure path. That matches announced reserves on the equilibrium path up to the lower support, but not when the seller raises the reserve above $\underline{r}$ and failure reveals bids above $\underline{r}$.

Claim 2. *The equilibrium lower support $\underline{r}^*$ is the argmax of $\Pi(\cdot)$ in (18).*

4.4 Proof of Claim 2

Proof. Let $\underline{r}^* \in \arg\max \Pi(\cdot)$. We show that any other lower support $\underline{r} \neq \underline{r}^*$ cannot be optimal.

Downward deviation. If $\underline{r} > \underline{r}^*$, stage-2 buyers keep the prior ψ, so (18) is exact and $\Pi(\underline{r}^*) > \Pi(\underline{r})$.

Upward deviation. If $\underline{r} < \underline{r}^*$, let $\tilde{\psi}(\beta(v)) \equiv \psi(1-\rho(\beta(v)))/(1-\psi\rho(\beta(v)))$ as in (5). The failure branch from announcing $\underline{r}^*$ is

$$
\begin{aligned}
&\delta N(c_H - c_L) \int_{c_L}^{\beta^{-1}(\underline{r})} 1 - \frac{F\left(\frac{c_H - \psi c_L}{1-\psi}\right)}{F(v)}\, dF(v)^N \\
&\quad + \delta N(c_H - c_L) \int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(\underline{r}^*)} 1 - \frac{F\left(\frac{c_H - \tilde{\psi}(\beta(v))c_L}{1-\tilde{\psi}(\beta(v))}\right)}{F(v)}\, dF(v)^N. \qquad (19)
\end{aligned}
$$

The success branch is the same as in (17) with $\underline{r}^*$ in place of $\underline{r}$. For $v \in [\beta^{-1}(\underline{r}), \beta^{-1}(\underline{r}^*)]$, $\tilde{\psi}(\beta(v)) < \psi$, so $F\left(\frac{c_H - \tilde{\psi}(\beta(v))c_L}{1-\tilde{\psi}(\beta(v))}\right) < F\left(\frac{c_H - \psi c_L}{1-\psi}\right)$ and the failure branch above exceeds the failure term in $\Pi(\underline{r}^*)$. Hence announcing $\underline{r}^*$ yields profit at least $\Pi(\underline{r}^*) > \Pi(\underline{r})$.

Conclusion. Either deviation is profitable, so $\underline{r}^*$ is the optimal lower support. $\square$

The low-cost seller's equilibrium payoff is $\Pi(\underline{r}^*)$. On the low branch, $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1-\delta\psi)$.

5 Low-Cost Seller: Optimal Support and Mixed Strategy

This section completes the low-cost seller's problem. Claim 2 pins the equilibrium lower support to the reserve that maximizes $\Pi(\cdot)$ in (18). We characterize that maximizer, extend payoffs to reserves above the lower support on the mixing path, and construct the reserve distribution from the resulting indifference condition.

5.1 Optimal lower support

The lower support $\underline{r}^*$ maximizes $\Pi(\underline{r})$ in (18). For an interior maximizer with $\underline{r}^* > \underline{r}_0$ from (14), the positive part in (18) is active. Imposing $\mathrm{d}\Pi(\underline{r})/\mathrm{d}\underline{r} = 0$ and rearranging yields the first-order condition

$$
\begin{aligned}
\underline{r}^* - c_L &= (1-\delta\psi)\, \frac{1 - F\left(\beta^{-1}(\underline{r}^*)\right)}{f\left(\beta^{-1}(\underline{r}^*)\right)} \qquad (20)\\
&\quad + \frac{1}{F(\underline{r}^*)^{N-1}} \Bigg[\delta(c_H - c_L)\, F\left(\beta^{-1}(\underline{r}^*)\right)^{N-2} \left(NF\left(\beta^{-1}(\underline{r}^*)\right) - (N-1)F\left(\frac{c_H - \psi c_L}{1-\psi}\right)\right) \\
&\quad - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} (s - c_L)\, dF(s)^{N-1} \Bigg].
\end{aligned}
$$

The derivation is in Appendix A.

5.2 Profit and indifference on the mixing path

In equilibrium the low-cost seller mixes a reserve $r \in [\underline{r}, \bar{r}]$ with cdf ρ, where $\underline{r}$ and $\bar{r}$ are the endogenous bounds of the support. If the realized reserve exceeds $\underline{r}$, stage 1 failure can occur with

$v_{\max}$ between $\beta^{-1}(\underline{r})$ and $\beta^{-1}(r)$; bids then reveal $\rho(\beta(v)) > 0$ and beliefs update via (5). On the failure path, beliefs split at $\beta^{-1}(\underline{r})$: the prior ψ applies below that threshold and $\tilde{\psi}(\beta(v))$ applies above it until $\beta^{-1}(r)$. Payoff from announcing $r > \underline{r}$ is

$$\begin{aligned}
\hat{\Pi}(r) = \delta N(c_H - c_L)\Big[\Big(&F\big(\beta^{-1}(\underline{r})\big) - F\Big(\frac{c_H - \psi c_L}{1-\psi}\Big)\Big)F\big(\beta^{-1}(\underline{r})\big)^{N-1} \\
&+ F\big(\beta^{-1}(r)\big)^N - F\big(\beta^{-1}(\underline{r})\big)^N - N\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} F\Big(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\Big) \\
&\qquad \times F(v)^{N-2}\, dF(v)\Big] \\
&+ N\Big\{(r - c_L)F(r)^{N-1}\big(1 - F(\beta^{-1}(r))\big) \\
&\quad + \int_{\beta^{-1}(r)}^{1}\int_{r}^{v}(s - c_L)\, dF(s)^{N-1}\, dF(v)\Big\}.
\end{aligned} \tag{21}$$

Appendix B derives the middle integral. When $r = \underline{r}$, the middle terms vanish and (21) coincides with (18).

On the equilibrium path $\underline{r} = \underline{r}^*$, so the seller is indifferent across the support:

$$\hat{\Pi}(r) = \Pi(\underline{r}^*) \qquad \forall\, r \in [\underline{r}, \bar{r}]. \tag{22}$$

5.3 Construction of the mixing distribution

Differentiating (22) on the low branch yields an ordinary differential equation for $h(r) \equiv \beta^{-1}(r)$ and the mixing cdf $\rho(r)$; the algebra is in Appendix C. On the low branch, $h(r)$ coincides with (9) at $b = r$:

$$h(r) = \frac{(1-\psi\rho(r))r - \delta\psi(1-\rho(r))c_L}{1 - \delta\psi - (1-\delta)\psi\rho(r)}, \tag{23}$$

with inverse

$$\rho(r) = \frac{-h(r) + r + \delta\psi(h(r) - c_L)}{\psi\big(-h(r) + r + \delta(h(r) - c_L)\big)}. \tag{24}$$

Indifference on the low branch is equivalent to

$$h'(r) = \frac{F(r)^{N-1}\big(1 - F(h(r))\big)}{f(h(r))} \times \mathcal{B}(r, h(r))^{-1}, \tag{25}$$

where

$$\begin{aligned}
\mathcal{B}(r,h) = (h - c_L)F(h)^{N-1} - &\int_r^h F(v)^{N-1}\, dv \\
&- \delta N(c_H - c_L)\left[F(h) - F\Big(\frac{(1-\psi\rho(r))c_H - \psi(1-\rho(r))c_L}{1-\psi}\Big)\right]F(h)^{N-2}.
\end{aligned} \tag{26}$$

Claim 3. *Suppose F and f are continuous on $[c_L, 1]$ with $f > 0$. Let $\underline{r}^*$ satisfy (20), and set*

$$h^* = \frac{\underline{r}^* - \delta\psi c_L}{1 - \delta\psi},$$

the inverse of the first bidding branch in (8). Then (25) admits a local solution $r \mapsto h(r)$ through $\big(\underline{r}^*, h^*\big)$.

Proof. At the lower end of the support, $\rho(\underline{r}^*) = 0$. Hence buyers use the prior belief ψ, bid according to $v - \delta\psi(v - c_L)$, and the inverse bid is the displayed h^*. Appendix C.3 shows that the denominator $\mathcal{B}(\underline{r}^*, h^*)$ is strictly positive. The right-hand side of (25) is therefore continuous around $\big(\underline{r}^*, h^*\big)$. Peano's theorem then gives a local solution. $\square$

A local solution is enough to define a mixed strategy on an interval $[\underline{r}^*, r_1]$: given h solving (25) on that interval, (24) pins down ρ, and (22) holds by construction on the low branch. For $r > \beta(v^*)$, $\beta^{-1}(r)$ is known in closed form and (43) in Appendix C extends ρ on the upper part of $[\underline{r}, \bar{r}]$. Together with $\bar{r} < r_H$ (Claim 1), this yields a mixed strategy equilibrium for the low-cost seller with payoff $\Pi(\underline{r}^*)$.

6 Upper Support and $\bar{r}$

At the top of the low-cost mixing support, $\rho(\bar{r}) = 1$. By (5), a stage 1 failure at bid $\bar{r}$ therefore leaves buyers with probability zero that the seller is low-cost: $\tilde{\psi}(\bar{r}) = 0$. By (1), a selected buyer offers c_H only if $v \geq c_H$.

6.1 Payoff at $r = \bar{r}$

Set $\underline{r} = \underline{r}^*$ on the equilibrium path and $r = \bar{r}$ in (21). Write $z^* \equiv \beta^{-1}(\underline{r}^*)$ and $z_{\bar{r}} \equiv \beta^{-1}(\bar{r})$. On the upper part of the support,

$$z_{\bar{r}} = \frac{\bar{r} - \delta c_H}{1 - \delta} \qquad \text{when } \bar{r} > \beta(v^*), \tag{27}$$

as in (8) and Appendix C.4. Then

$$\begin{aligned}
\hat{\Pi}(\bar{r}) = \delta N(c_H - c_L)\Big[&\Big(F(z^*) - F\Big(\frac{c_H - \psi c_L}{1-\psi}\Big)\Big)F(z^*)^{N-1} \\
&+ F(z_{\bar{r}})^N - F(z^*)^N - N\int_{z^*}^{z_{\bar{r}}} F\bigg(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\bigg) F(v)^{N-2}\, dF(v)\Big] \\
&+ N\Big\{(\bar{r} - c_L)F(\bar{r})^{N-1}\big(1 - F(z_{\bar{r}})\big) \\
&\quad + \int_{z_{\bar{r}}}^{1}\int_{\bar{r}}^{v}(s - c_L)\, dF(s)^{N-1}\, dF(v)\Big\}.
\end{aligned} \tag{28}$$

This is (21) evaluated at the upper support. The ψ block is the same as in (18) because it depends only on $\underline{r}^*$, not on $\bar{r}$. The remaining terms are the incremental failure and success payoffs from raising the reserve from $\underline{r}^*$ to $\bar{r}$.

At the upper endpoint of the failure integral, $\rho(\beta(z_{\bar{r}})) = \rho(\bar{r}) = 1$, so the integrand cutoff equals c_H:

$$\frac{(1-\psi\rho(\bar{r}))c_H - \psi(1-\rho(\bar{r}))c_L}{1-\psi} = c_H.$$

On the equilibrium path, indifference gives

$$\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*) = \hat{\Pi}(\underline{r}^*), \tag{29}$$

where the second equality uses (21) at $r = \underline{r}^* = \underline{r}$.

6.2 Proof of Claim 1

Proof. The low-cost seller mixes on $[\underline{r}^*, \bar{r}]$ with $\rho(\bar{r}) = 1$. At bid $\bar{r}$, (5) gives $\tilde{\psi}(\bar{r}) = 0$, so buyers assign probability zero to the low type after failure at the top of the support. If $\bar{r} \geq r_H$, the high-cost seller's optimal pure reserve would lie inside (or below) the low-cost mixing support.

Consider the high-cost seller's problem in (3). Its optimal reserve r_H satisfies (4) and is strictly larger than the reserve that maximizes the low-cost *pure*-strategy profit $\Pi(\cdot)$ in (18): a high-cost seller faces a higher stage 2 cutoff and a steeper stage 1 bidding schedule $\beta(v) = v - \delta(v - c_H)$ than the low-cost branch $\beta(v) = v - \delta\psi(v - c_L)$ with $\psi < 1$. Hence $r_H > \arg\max_r \Pi(r) \geq \underline{r}^*$.

On the equilibrium path, the low-cost seller is indifferent across $[\underline{r}^*, \bar{r}]$, so $\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*)$ by (29). The payoff $\hat{\Pi}(\bar{r})$ in (28) is built from the low-cost bidding rule up to $z_{\bar{r}}$ in (27). If $\bar{r} \geq r_H$, the high-cost seller could meet reserve r_H while bidders above $\bar{r}$ already treat the seller as high-cost (Section 2), so the high-cost type's optimal separation reserve would not exceed the top of the low-cost support. That contradicts $r_H > \arg\max \Pi(\cdot)$ and the definition of $\bar{r}$ as the upper endpoint of the low-cost mixing support with $\rho(\bar{r}) = 1$.

Therefore $\bar{r} < r_H$. □

7 Case of Public Reserve Price

If the equilibrium is separating, the low-cost seller sets a reserve price given by

$$r_L - c_L = (1-\delta)\frac{1 - F\left(\dfrac{r_L - \delta c_L}{1-\delta}\right)}{f\left(\dfrac{r_L - \delta c_L}{1-\delta}\right)} - \frac{1}{F(r_L)^{N-1}} \int_{r_L}^{(r_L - \delta c_L)/(1-\delta)} (v - c_L)\, dF(v)^{N-1}. \tag{30}$$

whereas the high-cost seller sets reserve price weakly above r_H, solution to

$$r_H - c_H = (1-\delta)\frac{1 - F\left(\dfrac{r_H - \delta c_H}{1-\delta}\right)}{f\left(\dfrac{r_H - \delta c_H}{1-\delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{(r_H - \delta c_H)/(1-\delta)} (v - c_H)\, dF(v)^{N-1}, \tag{31}$$

that is enough to deter low-type mimicking.

A FOC for $\underline{r}^*$

This appendix derives (20) from (18) on the interior region where the positive part in (15) is active.

A.1 Derivative of the failure-region term

$$\begin{aligned}
&\frac{d}{d\underline{r}}\left[\delta N(c_H-c_L)\times\left(F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-1}\right]\\
&=\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-1}\\
&\quad+(N-1)\frac{1}{1-\delta\psi}\left(F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)\\
&=\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2}\left[NF\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-(N-1)F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right]. \qquad (32)
\end{aligned}$$

A.2 Simplifying the double integral

$$\begin{aligned}
&\int_{\beta^{-1}(\underline{r})}^{1}\int_{\underline{r}}^{v}(s-c_L)\,dF(s)^{N-1}\,dF(v)\\
&=\left[F(v)\int_{\underline{r}}^{v}(s-c_L)\,dF(s)^{N-1}\right]_{\beta^{-1}(\underline{r})}^{1}-\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}\\
&=\int_{\underline{r}}^{1}(s-c_L)\,dF(s)^{N-1}-F\big(\beta^{-1}(\underline{r})\big)\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}-\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}. \qquad (33)
\end{aligned}$$

Derivative of the first term.

$$\frac{d}{d\underline{r}}\int_{\underline{r}}^{1}(s-c_L)\,dF(s)^{N-1}=-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r}). \qquad (34)$$

Derivative of the second term.

$$\begin{aligned}
&\frac{d}{d\underline{r}}\left(F\big(\beta^{-1}(\underline{r})\big)\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\right)\\
&=f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\\
&\quad+(N-1)F\big(\beta^{-1}(\underline{r})\big)\Big[\big(\beta^{-1}(\underline{r})-c_L\big)F\big(\beta^{-1}(\underline{r})\big)^{N-2}f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}-(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\Big]. \qquad (35)
\end{aligned}$$

Derivative of the third term.

$$\frac{d}{d\underline{r}}\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}=-\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}(N-1)\big(\beta^{-1}(\underline{r})-c_L\big)F\big(\beta^{-1}(\underline{r})\big)^{N-1}f\big(\beta^{-1}(\underline{r})\big). \qquad (36)$$

The derivative of the double integral is therefore

$$-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big)-f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}. \qquad (37)$$

Derivative of the bracket term.

$$
\begin{aligned}
&\frac{d}{d\underline{r}}\left[(\underline{r}-c_L)F(\underline{r})^{N-1}\big(1-F(\beta^{-1}(\underline{r}))\big)\right] \\
&=\begin{bmatrix} F(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big) \\ \quad +(N-1)(\underline{r}-c_L)\big(1-F(\beta^{-1}(\underline{r}))\big)f(\underline{r}) \end{bmatrix} F(\underline{r})^{N-2}. \\
&\qquad -(\underline{r}-c_L)F(\underline{r})f\big(\beta^{-1}(\underline{r})\big)\frac{1}{1-\delta\psi}
\end{aligned}
\tag{38}
$$

A.3 Full FOC and simplification

Taken together, $\mathrm{d}\Pi(\underline{r})/\mathrm{d}\underline{r}=0$ is

$$
\begin{aligned}
0 &= \delta N(c_H-c_L)\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2} \\
&\quad\times\left[NF\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-(N-1)F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right] \\
&\quad+N\begin{bmatrix} F(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big) \\ \quad +(N-1)(\underline{r}-c_L)\big(1-F(\beta^{-1}(\underline{r}))\big)f(\underline{r}) \end{bmatrix} F(\underline{r})^{N-2} \\
&\qquad -(\underline{r}-c_L)F(\underline{r})f\big(\beta^{-1}(\underline{r})\big)\frac{1}{1-\delta\psi} \\
&\quad+N\Big[-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big) \\
&\qquad -f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\Big].
\end{aligned}
\tag{39}
$$

The $(N-1)(\underline{r}-c_L)$ terms from the bracket and double-integral derivatives cancel. Solving the remaining expression for $\underline{r}-c_L$ gives (20). The outer factor N in the failure derivative and the outer factor N in the success derivative cancel in this final step.

B Profit for $r>\underline{r}$

Write $v_\psi \equiv (c_H-\psi c_L)/(1-\psi)$ and $v_{\tilde{\psi}}(v)\equiv\big((1-\psi\rho(\beta(v)))c_H-\psi(1-\rho(\beta(v)))c_L\big)/(1-\psi)$. The stage-2 failure block in (21) is

$$
\begin{aligned}
&\delta N(c_H-c_L)\big(F\big(\beta^{-1}(\underline{r})\big)-F(v_\psi)\big)\,F\big(\beta^{-1}(\underline{r})\big)^{N-1} \\
&\quad+\delta N(c_H-c_L)\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}\left(1-\frac{F\big(v_{\tilde{\psi}}(v)\big)}{F(v)}\right)dF(v)^N,
\end{aligned}
\tag{40}
$$

plus the stage-1 success terms in (21). The first line matches (15) at $r=\underline{r}$. Since $\int(1-F_{\tilde{\psi}}/F)\,dF^N = N\int(F-F_{\tilde{\psi}})F^{N-2}\,dF$,

$$
\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}\left(1-\frac{F_{\tilde{\psi}}}{F}\right)dF^N = F\big(\beta^{-1}(r)\big)^N - F\big(\beta^{-1}(\underline{r})\big)^N - N\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}F_{\tilde{\psi}}\,F^{N-2}\,dF,
$$

which yields the middle terms in (21).

C Mixed-strategy indifference and the ODE

C.1 Incremental indifference

On $[\underline{r}, \bar{r}]$, equilibrium requires $\hat{\Pi}(r) = \Pi(\underline{r}^*)$. The ψ block in (21) does not depend on r; with $\underline{r} = \underline{r}^*$ on path,

$$\begin{aligned}
&\delta N(c_H - c_L)\Bigg[F\big(\beta^{-1}(r)\big)^N - F\big(\beta^{-1}(\underline{r}^*)\big)^N - N\int_{\beta^{-1}(\underline{r}^*)}^{\beta^{-1}(r)} F\left(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\right) F(v)^{N-2}\, dF(v)\Bigg] \\
&\quad + N\left\{(r - c_L)F(r)^{N-1}\big(1 - F(\beta^{-1}(r))\big) + \int_{\beta^{-1}(r)}^{1}\int_{r}^{v}(s - c_L)\, dF(s)^{N-1}\, dF(v)\right\} \\
&= N\left\{(\underline{r}^* - c_L)F(\underline{r}^*)^{N-1}\big(1 - F(\beta^{-1}(\underline{r}^*))\big) + \int_{\beta^{-1}(\underline{r}^*)}^{1}\int_{\underline{r}^*}^{v}(s - c_L)\, dF(s)^{N-1}\, dF(v)\right\}. \qquad (41)
\end{aligned}$$

C.2 Derivative and the ODE

On the low branch, $\beta^{-1}(r) = h(r)$. Differentiating (41) in r gives

$$\begin{aligned}
0 &= \delta N(c_H - c_L)\left[F\big(h(r)\big)^{N-1} f\big(h(r)\big)\, h'(r) - F\left(\frac{(1-\psi\rho(r))c_H - \psi(1-\rho(r))c_L}{1-\psi}\right) F\big(h(r)\big)^{N-2} f\big(h(r)\big)\, h'(r)\right] \\
&\quad + F(r)^{N-1}\big(1 - F(h(r))\big) - (r - c_L)F(r)^{N-1} f\big(h(r)\big)\, h'(r) \\
&\quad - f\big(h(r)\big)\, h'(r)\int_{r}^{h(r)}(s - c_L)\, dF(s)^{N-1}. \qquad (42)
\end{aligned}$$

Collecting terms with $h'(r)$ in (42) and solving gives (25)–(26).

C.3 Initial denominator

At the lower support, $\rho(\underline{r}^*) = 0$, so buyers keep the prior belief ψ after failure. The first bidding branch in (8) therefore gives

$$\beta^{-1}(\underline{r}^*) = \frac{\underline{r}^* - \delta\psi c_L}{1 - \delta\psi}.$$

The stage-2 cutoff in (26) also reduces to $(c_H - \psi c_L)/(1-\psi)$. To evaluate the ODE denominator at the initial point, first use

$$\begin{aligned}
&\int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)}(s - c_L)\, dF(s)^{N-1} \\
&\qquad = \big(\beta^{-1}(\underline{r}^*) - c_L\big)F\big(\beta^{-1}(\underline{r}^*)\big)^{N-1} - (\underline{r}^* - c_L)F(\underline{r}^*)^{N-1} - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} F(v)^{N-1}\, dv.
\end{aligned}$$

Substituting this identity into (26) and using (20) gives

$$\begin{aligned}
&\mathcal{B}\big(\underline{r}^*, \beta^{-1}(\underline{r}^*)\big) \\
&\quad = (1 - \delta\psi)\,\frac{1 - F\big(\beta^{-1}(\underline{r}^*)\big)}{f\big(\beta^{-1}(\underline{r}^*)\big)}\, F(\underline{r}^*)^{N-1} \\
&\qquad + \delta(c_H - c_L)F\left(\frac{c_H - \psi c_L}{1-\psi}\right) F\big(\beta^{-1}(\underline{r}^*)\big)^{N-2}.
\end{aligned}$$

This is the initial denominator used in Claim 3. It is strictly positive whenever $F(\beta^{-1}(\underline{r}^*)) < 1$.

C.4 High reserves

For $r > \beta(v^*)$, $\beta^{-1}(r) = (r - \delta c_H)/(1-\delta)$ and (42) becomes

$$\begin{aligned}
0 = {} & \delta N(c_H - c_L)\left(F\left(\frac{r-\delta c_H}{1-\delta}\right) - F\left(\frac{(1-\psi\rho(r))c_H - \psi(1-\rho(r))c_L}{1-\psi}\right)\right) F\left(\frac{r-\delta c_H}{1-\delta}\right)^{N-2} f\left(\frac{r-\delta c_H}{1-\delta}\right)\frac{1}{1-\delta} \\
& + F(r)^{N-1}\left(1 - F\left(\frac{r-\delta c_H}{1-\delta}\right)\right) \\
& - \frac{1}{1-\delta} f\left(\frac{r-\delta c_H}{1-\delta}\right)\left[\left(\frac{r-\delta c_H}{1-\delta} - c_L\right) F\left(\frac{r-\delta c_H}{1-\delta}\right)^{N-1} - \int_r^{(r-\delta c_H)/(1-\delta)} F(v)^{N-1}\, dv\right],
\end{aligned} \tag{43}$$

which pins down $\rho(r)$ on the upper part of the support.